# Corporate Sustainability Reporting & ESG Analytics

## Methodology Note - Independent Student Case Study

GreenCore Technologies GmbH is fictional. All operational data is synthetic. This project demonstrates data-quality, analytics and selected sustainability-reporting concepts and is not an ESRS/CSRD compliance assessment.

## 1. Reporting scope

Six simulated manufacturing sites in Europe and Asia, with monthly reporting from January 2023 through December 2025. The validated dataset contains 216 unique site-month records.

## 2. Environmental metrics

Scope 1 and Scope 2 GHG emissions, electricity, renewable electricity, natural gas, water withdrawal, total waste, recycled waste, hazardous waste, production volume and related intensity indicators.

## 3. Data-quality controls

| Control | Purpose |
| --- | --- |
| Missing required values | Identify incomplete submissions. |
| Negative values | Flag invalid negative environmental measurements. |
| Duplicates | Retain one site-month submission using latest-submission rule. |
| Renewable <= electricity | Prevent impossible energy relationships. |
| Recycled <= total waste | Prevent impossible waste relationships. |
| Production > 0 | Protect intensity calculations. |
| Unit standardization | Convert inconsistent energy units to MWh. |
| Master-data standardization | Align country/site values to site master. |

## 4. Audit approach

The raw dataset has 219 rows because three site-months were duplicated. The validation log documents 15 issues. Corrected values represent simulated site resubmission or master-data confirmation.

## 5. KPI logic

| KPI | Calculation |
| --- | --- |
| Total GHG | Scope 1 + Scope 2 |
| Renewable Share | Renewable electricity / total electricity |
| Recycling Rate | Recycled waste / total waste |
| Emissions Intensity | Total GHG / production x 1,000 |
| Water Intensity | Water withdrawal / production x 1,000 |

## 6. Reporting context

Selected educational mappings are used for ESRS E1 (Climate Change), E3 (Water) and E5 (Resource Use & Circular Economy). The European Commission adopted revised ESRS on 3 July 2026.

European Commission:
https://finance.ec.europa.eu/news/commission-adopts-revised-sustainability-reporting-standards-2026-07-03_en

EFRAG: https://www.efrag.org/en/sustainability-reporting

## 7. 2025 outputs

| Metric | 2025 result |
|---|---|
| Total GHG | 47,613.3 tCO2e |
| GHG change vs 2024 | -10.1% |
| Renewable share | 66.8% |
| Recycling rate | 85.3% |
| Water withdrawal | 3,142,731 m3 |